Article Recommender System

DS300.Q11 - Final Project

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DS300 Final Project Report

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Motivation: Why News Recommendation?

In the digital era, news platforms face **Information Overload**:

- **User Experience:** Navigating thousands of daily articles is overwhelming.
- **Retention:** Personalized content increases user engagement and platform stickiness.
- **Monetization:** Better engagement leads to higher ad revenue and subscriptions.

Target: Build a system that understands both *Content semantics* and *User behavior*.

Research Objectives

Our goal is to develop a state-of-the-art recommendation engine for Vietnamese news:

- **Hybrid Logic:** Combine Collaborative Filtering (CF) with Content-Based (CB) techniques.
- **Graph Learning:** Leverage GNNs to model complex user-article interactions.
- **Multimodal Fusion:** Incorporate Social (replies) and Temporal (time-decay) signals.
- **Vietnamese NLP:** Optimize PhoBERT embeddings for news semantics.

Problem Statement

Challenge: How to recommend relevant news articles to users?

Key Challenges:

1. **Cold Start:** New users/articles have no interaction history
2. **Sparse Data:** Most users read very few articles (99.9% sparsity)
3. **Vietnamese Language:** Limited NLP resources for Vietnamese text
4. **Dynamic Content:** News articles are time-sensitive

Our Approach: Graph-based Contrastive Learning with PhoBERT embeddings

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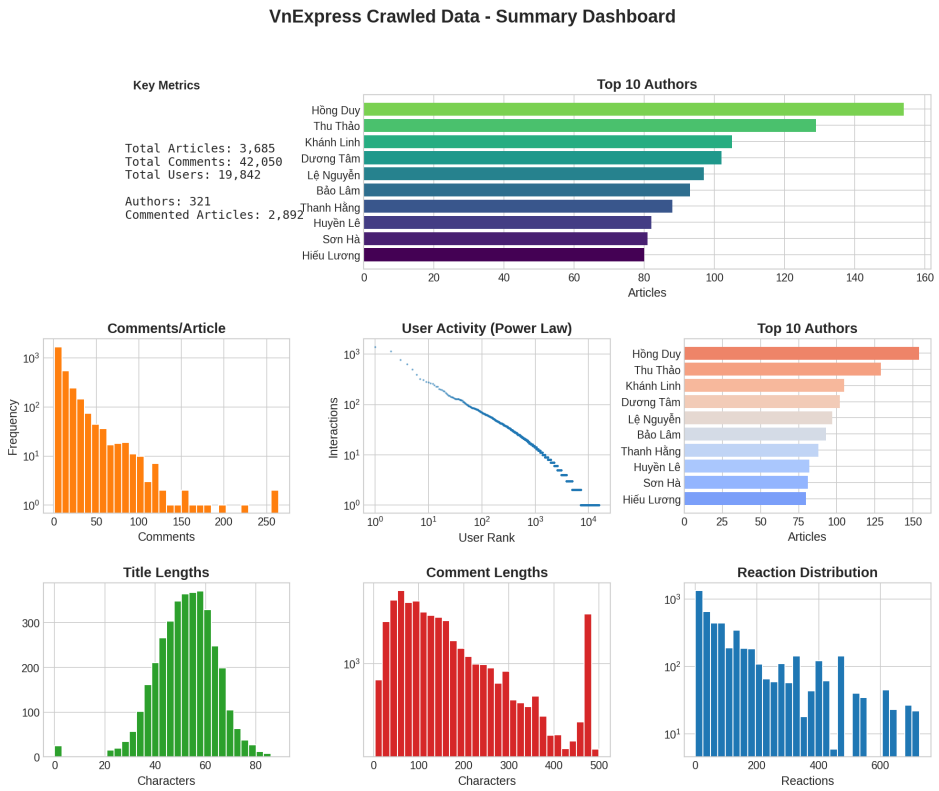
Dataset Overview (VnExpress)

Statistics:

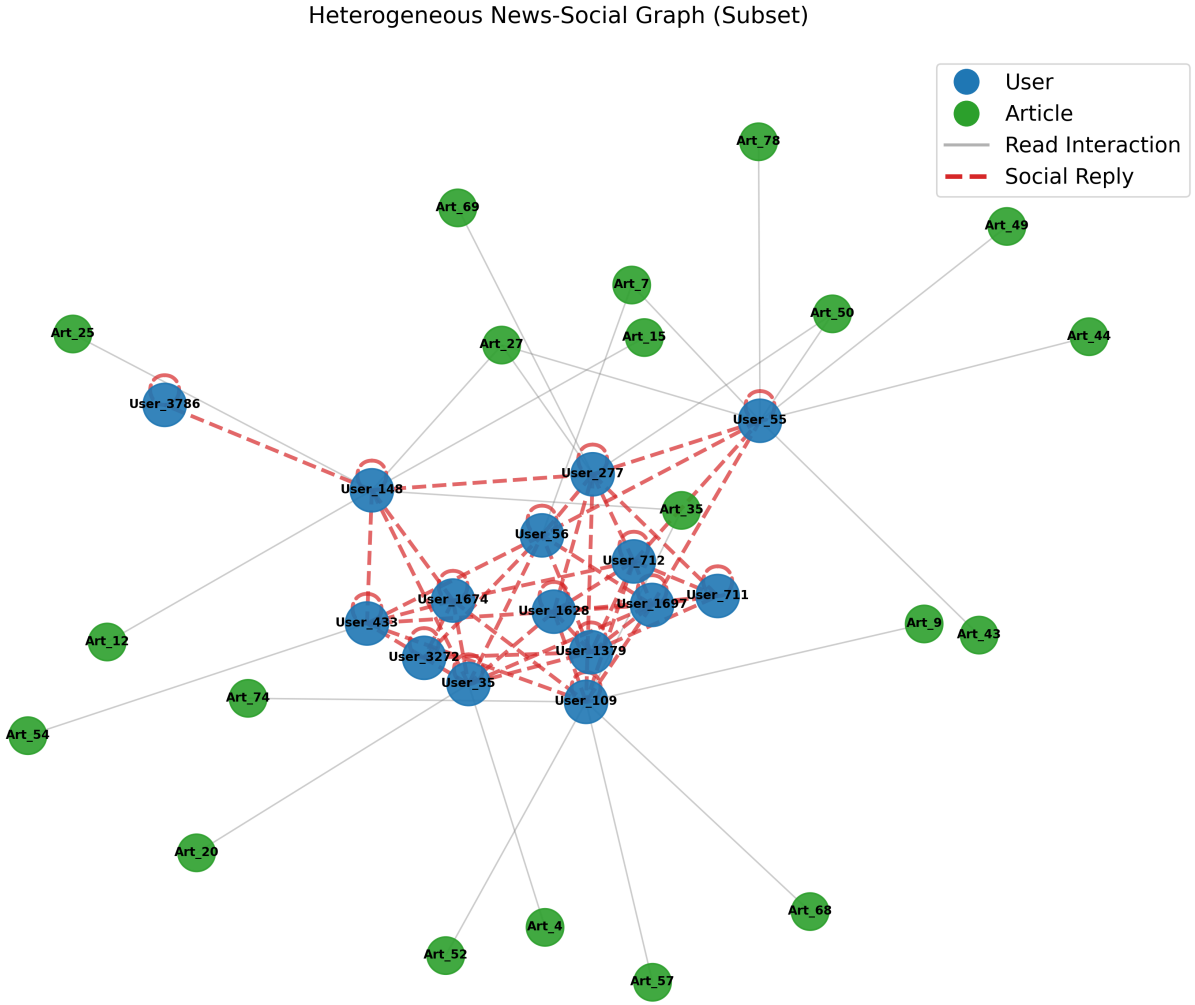
- **Users:** 13,117
- **Articles:** 3,645
- **Interactions:** 38,505
- **Sparsity:** 99.92%

The dataset is highly sparse, typical of news domains where users read only a tiny fraction of daily content.



Graph Topology

Visualizing the User-Article-Comment bipartite graph structure.



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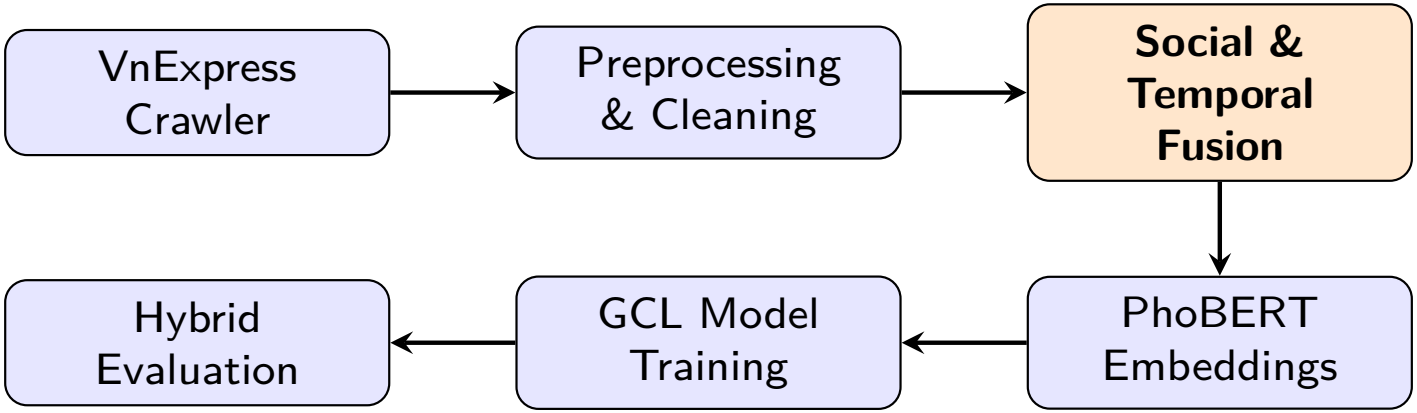
Current SOTA Approaches

We benchmark against three main categories of modern Recommender Systems:

1. **Graph Neural Networks (GNNs):**
   - **NGCF (2019):** First to encode collaborative signal into embeddings via graph propagation. High computational cost.
   - **LightGCN (2020):** Simplified GCN by removing non-linearities. Current industry standard for efficiency.
2. **Contrastive Learning (CL):**
   - **SimGCL (2022):** Augments graph structure with noise to learn robust representations. Handles sparsity better than pure GNNs.
3. **Content-Based & Hybrid:**
   - **PhoBERT:** State-of-the-art BERT model for Vietnamese, capturing deep semantic text features.

System Architecture



Pipeline: Crawl → Clean → Build Graph → Generate Embeddings → Train → Evaluate

Strategic Why: The Hybrid Synergy

Why combine **Content-Based** with **Graph Learning**?

- **Content-Based (NLP):**
  - Solves the **Cold-Start** problem (recommends new articles with 0 interactions).
  - Captures semantic similarity (PhoBERT).
- **Graph Learning (GNN):**
  - Captures **Collaborative Signals** (Users who like X also like Y).
  - Exploits **Social Signals** (Reply networks).

Hybrid Outcome

Content-Based provides "Freshness" while Graph Learning provides "Depth" and "Discovery".

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VnExpress Data Crawling

Data Source: VnExpress.net (largest Vietnamese news portal)

Categories Crawled:

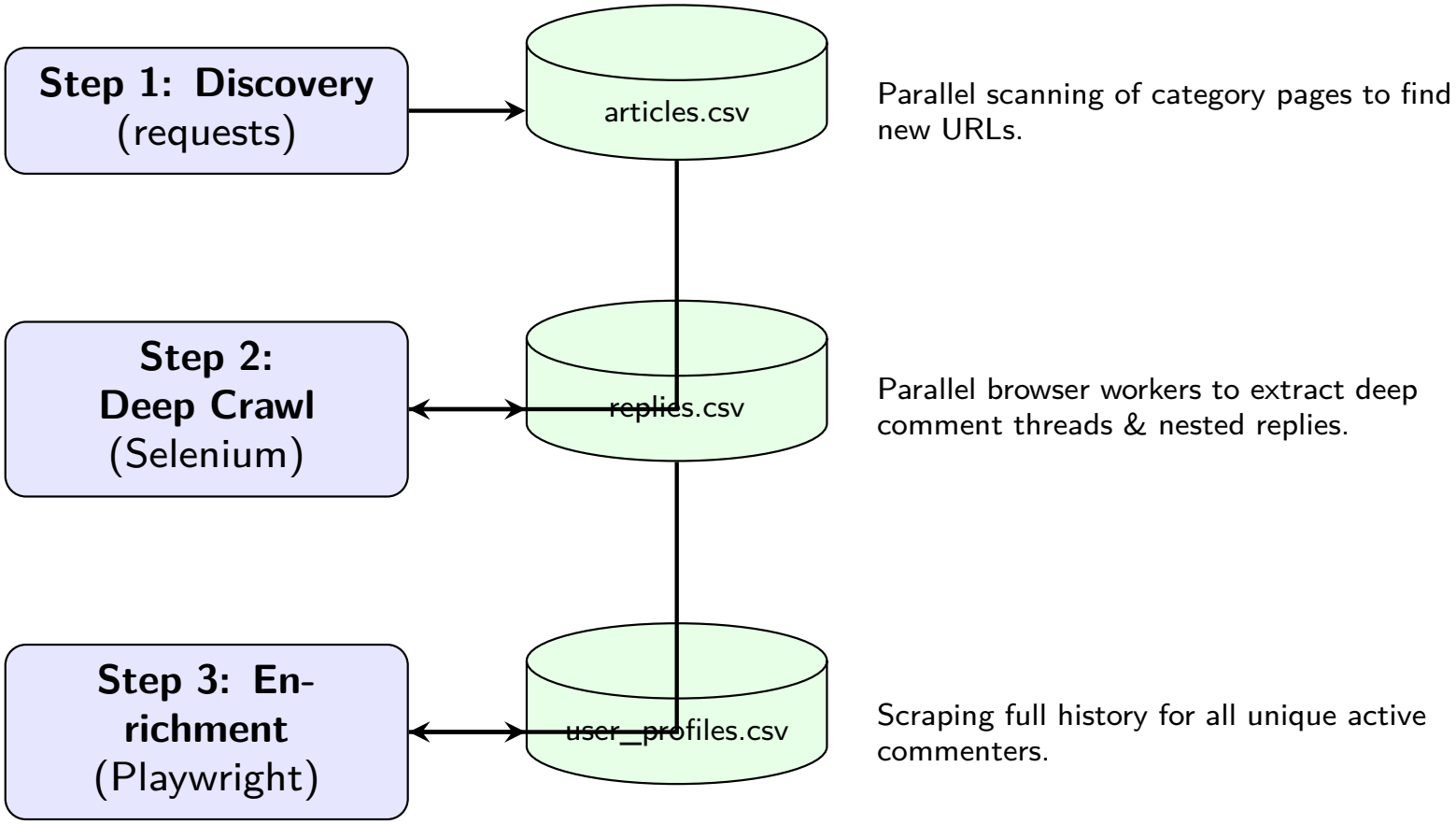
- 'Thế giới'
- 'Giáo dục'
- 'Thể thao'
- 'Thời sự'
- 'Kinh doanh'
- 'Khoa học công nghệ'

Collected Data:

- Articles: title, content, author, publish date
- Metadata: category, tags (per article)
- Comments: user ID, comment text, reactions, replies
- User profiles: user ID, username, join date

Data Collection Pipeline: 3-Step Workflow

Our crawler implements a resilient 3-stage architecture to handle high-volume news data:



Pipeline Orchestration & Efficiency

- **Parallelism:** `Step 2` (Selenium) is resource-intensive. We split the workload across N parallel browser instances using `multiprocessing`.
- **Resiliency:**
  - **Caching:** Avoids redundant HTTP requests for static content.
  - **Checkpointing:** Atomic writes to CSV allow resuming from the last article in case of failure.
  - **Headless Execution:** Minimizes resource overhead in production environments.
- **Tool Stack:**
  - **Requests:** High-speed metadata extraction.
  - **Selenium:** Handling complex JS (Load More buttons).
  - **Playwright:** Modern, fast engine for user history scraping.

Dataset Overview: High-Level Statistics

Raw Behavioral Data:

- **3,685** Articles
- **42,050** Comments
- **6** News Categories
- **19,842** User Profiles

Graph Interaction Set (G1-G3):

- **13,117** Unique Users
- **3,645** Active Articles
- **38,505** Total Interactions
- **Sparsity: 99.919%**

Data Sample: Interaction & Content

1. **Content Sample (Article)**
"Chinh phục loạt 'ông lớn' công nghệ sau hơn 500 lần bị từ chối"
→ **Tags:** [Meta, Amazon, Apple, Tesla, kĩ sư công nghệ]

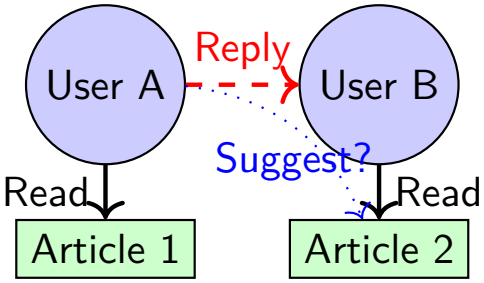
2. **Social Sample (Comment/Reply)**
   - **User A:** "Nếu giỏi thì cố gắng mua lại mấy công ty này..."
   - **User B:** (Reply to A) "Mua lại Meta chắc cần nghìn tỷ USD..."
   ⇒ Social Connection detected

Feature Mapping

- **Article** → PhoBERT Embedding ($d = 64$) → Node Feature.
- **User** → Interaction aggregation → Node Embedding.
- **Reply** → User-User Edge in the Heterogeneous Graph.

Core Innovation: Social Fusion

Idea: Use the "Reply Network" to bridge gaps between users.



The Logic:

- **Direct:** User A likes Article 1.
- **Social:** User A replies to User B $\rightarrow$ *Shared Interest*.
- **Inference:** User A is likely to enjoy Article 2 (read by B).

Impact on Sparsity

By adding User-User edges, the GNN can propagate "interests" even if User A has only read 1 article.

The Sparsity Challenge: Mathematical Proof

Definition:
Sparsity measures the lack of interactions in the potential link space.

The Formula

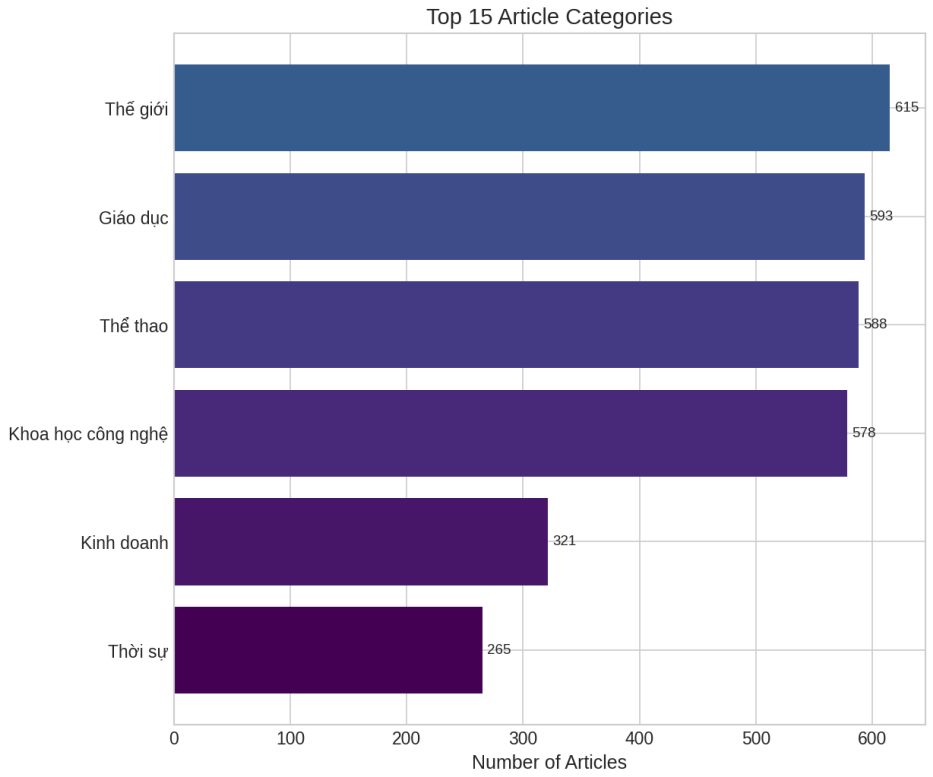
$$\text{Sparsity} = 1 - \frac{|\mathcal{E}_{obs}|}{|\mathcal{V}_u| \times |\mathcal{V}_i|}$$

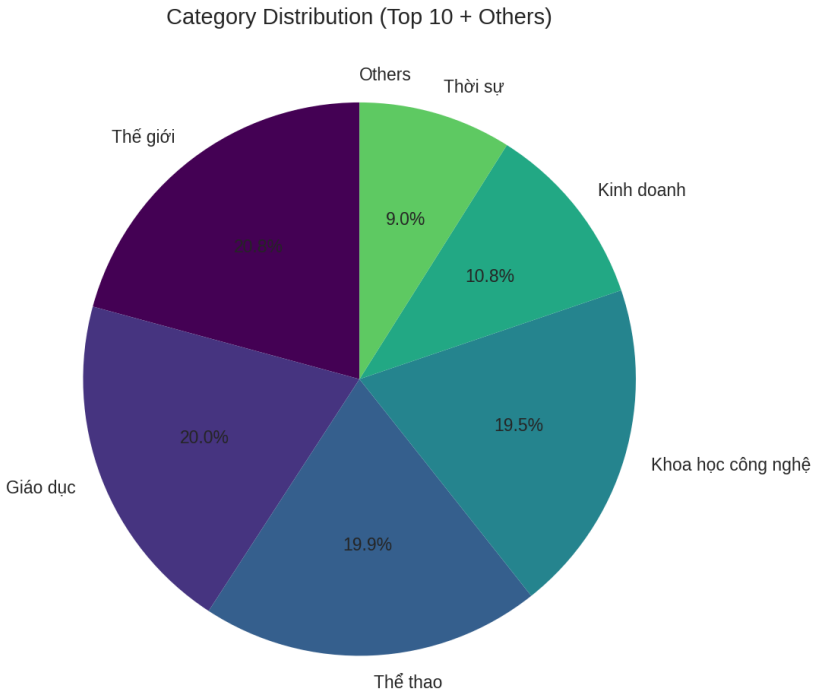
VnExpress Graph Calculation:

- Interactions ($|\mathcal{E}_{obs}|$): **38,505**
- Active Users ($|\mathcal{V}_u|$): **13,117**
- Active Articles ($|\mathcal{V}_i|$): **3,645**
- Total Possible Edges: $13,117 \times 3,645 = \mathbf{47,811,465}$

$$\Rightarrow \text{Sparsity} = 1 - \frac{38,505}{47,811,465} = \mathbf{99.919\%}$$

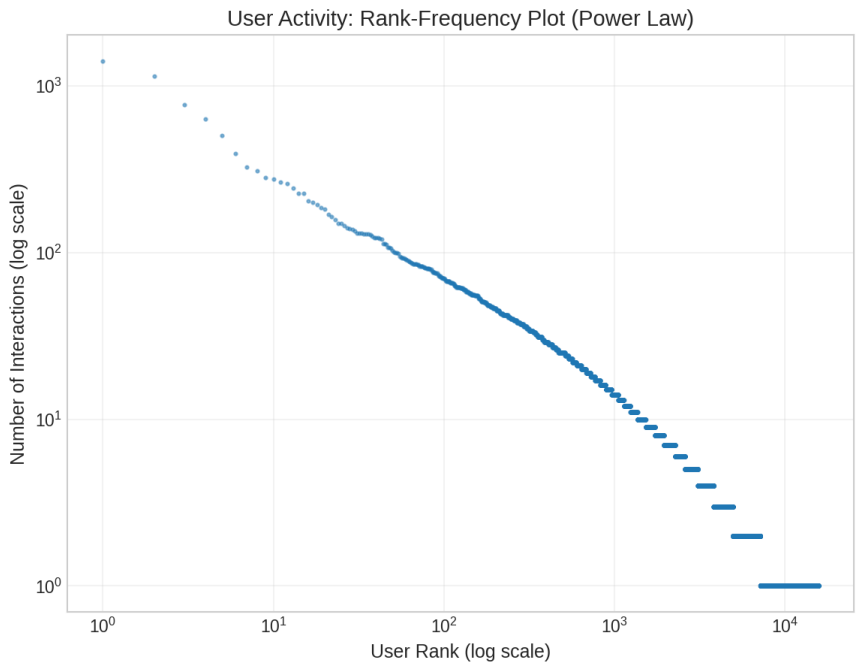
Category Distribution (VnExpress)

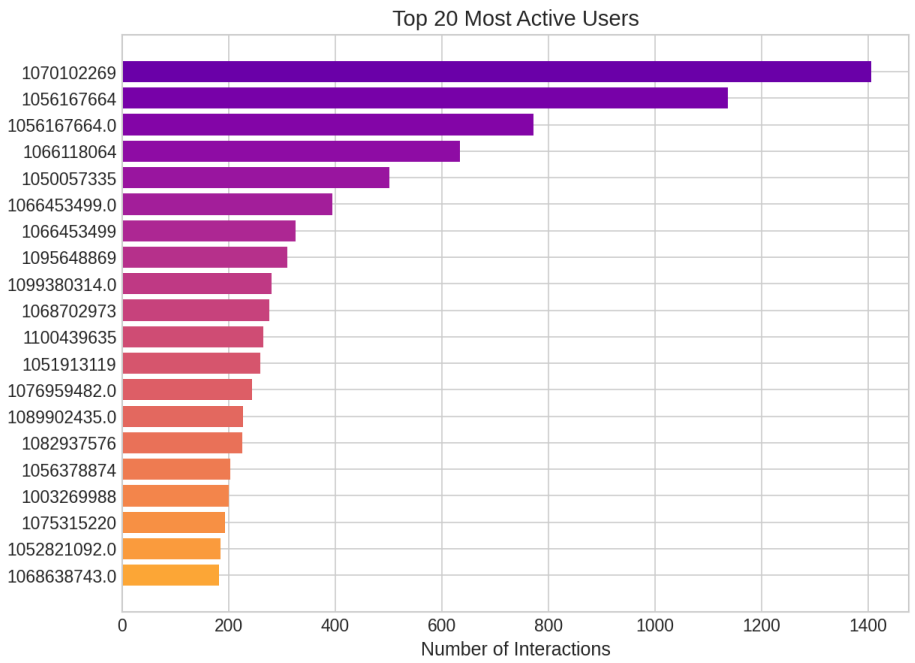




- **Thế giới** and **Thời sự** are the dominant categories.
- Skewed distribution reflects editorial priorities.

User Engagement Patterns (1/2)

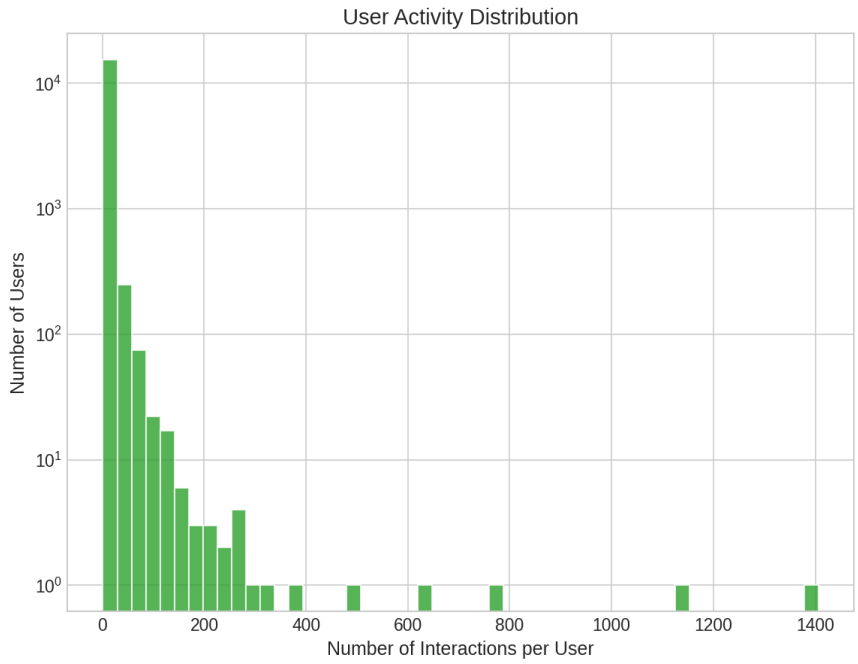


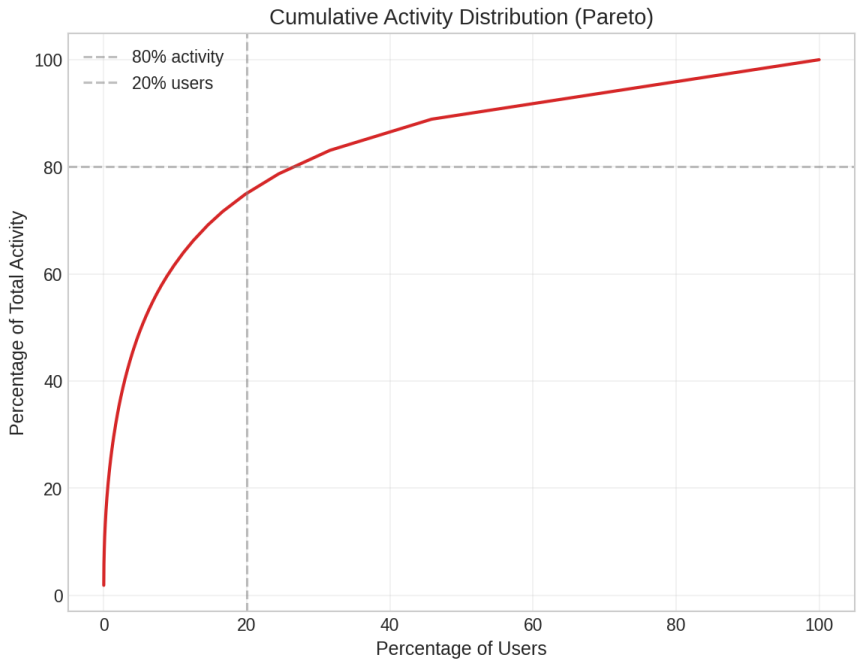


Distribution Analysis

Article popularity follows Zipf's law. Top 10 users account for 15% of all interactions.

User Engagement Patterns (2/2)

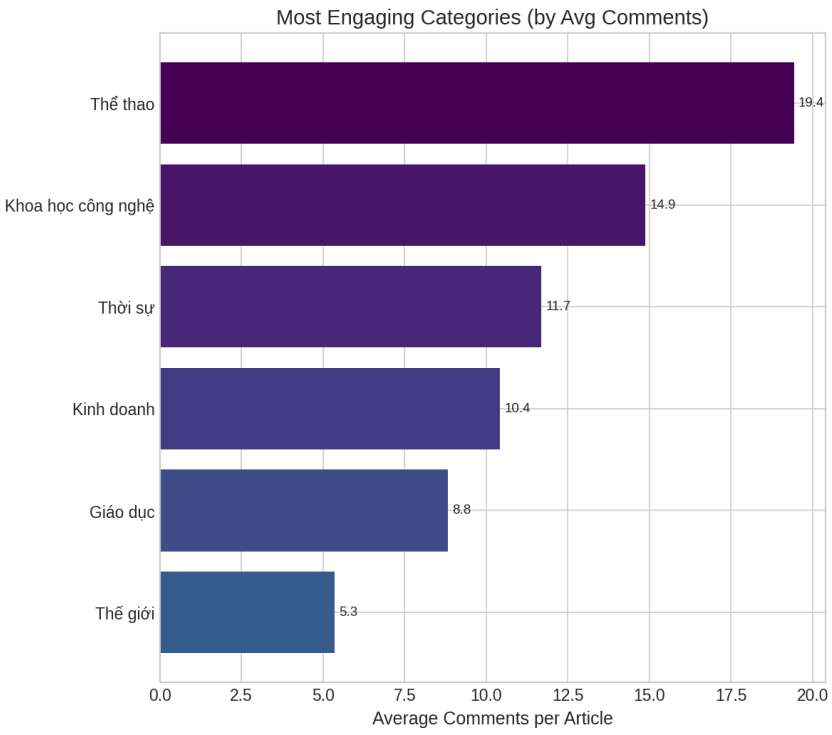


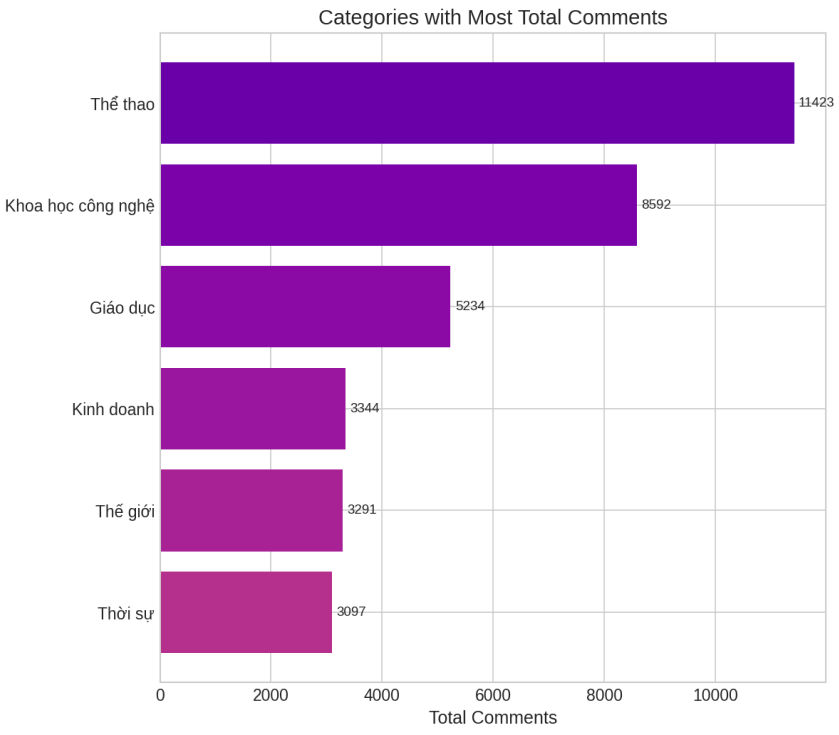


Power Law Observation

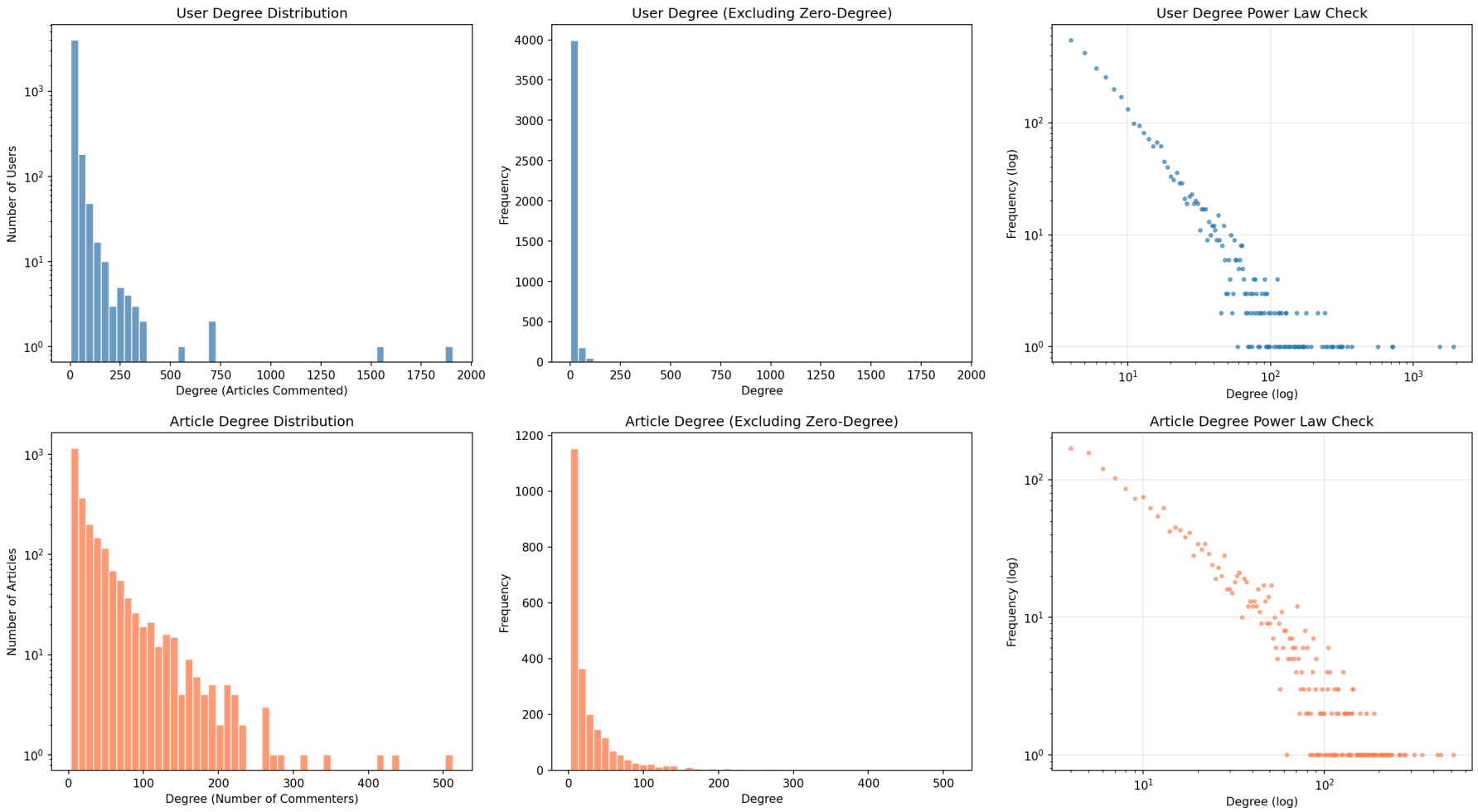
Long-tail distribution: 80% of users have fewer than 10 interactions.

Engagement by Category



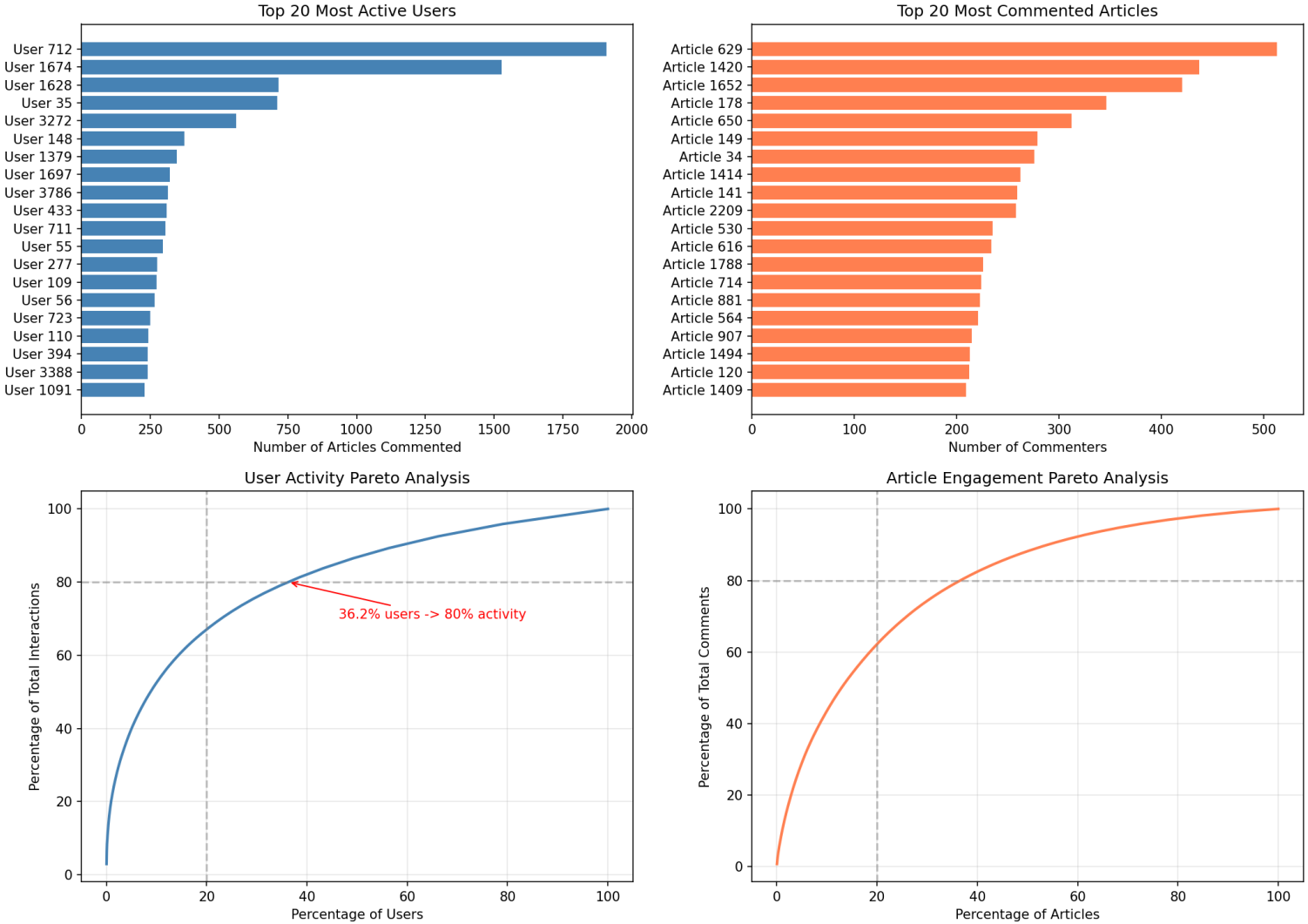


Graph Degree Distribution

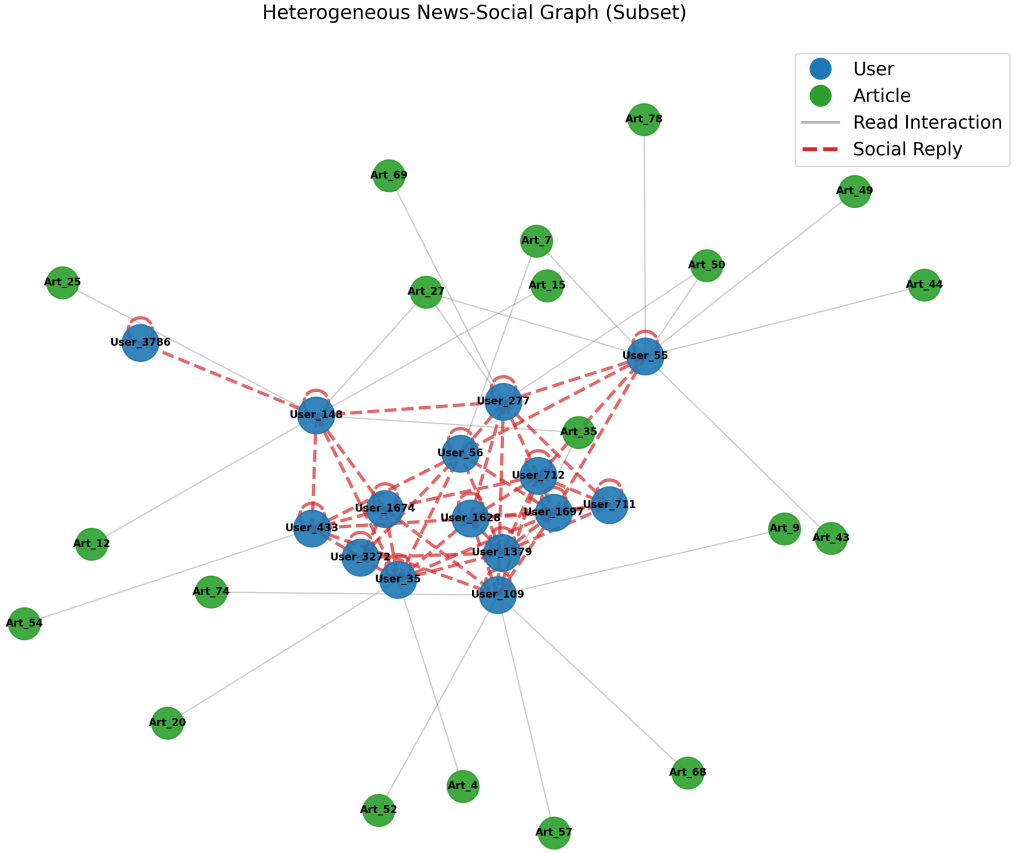


- User degrees peak at very low values (high cold-start risk).
- Article degrees are more distributed but still sparse.

Graph Connectivity & Density



Graph Topology: A Heterogeneous Network



- **Blue Nodes:** Active Users
- **Green Nodes:** News Articles
- **Dashed Edges:** Social Reply interactions

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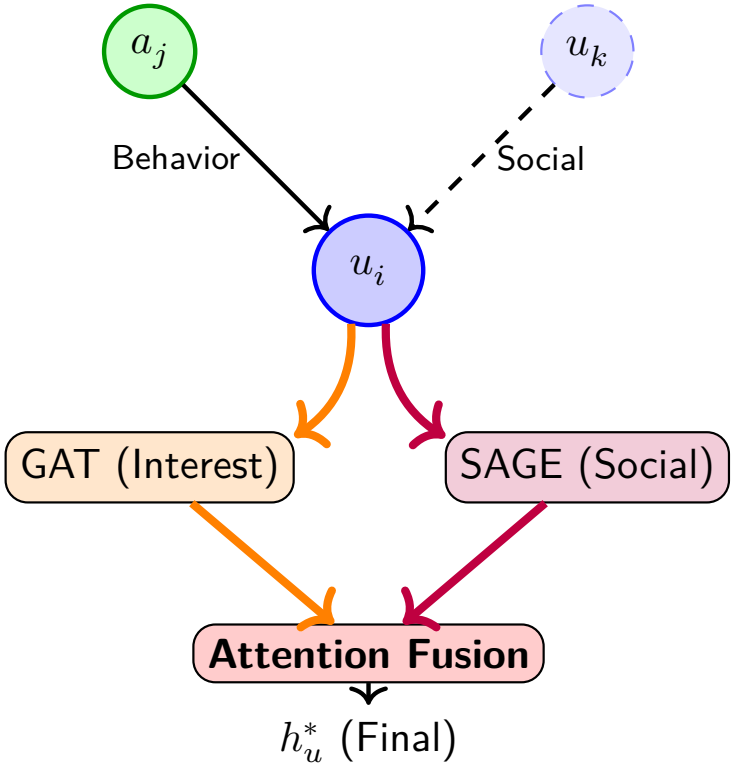
MA-HGN: Multi-Aspect Heterogeneous GNN



Core Innovation: Dual-path Propagation

- **Path 1 (Interest):** history ($\mathcal{G}_{u-a}$) via **GATv2**.
- **Path 2 (Social):** friends ($\mathcal{G}_{u-u}$) via **GraphSAGE**.
- **Adaptive Fusion:** balances signal importance:

$$h_u = \alpha \cdot h_{int} + (1 - \alpha) \cdot h_{soc}$$

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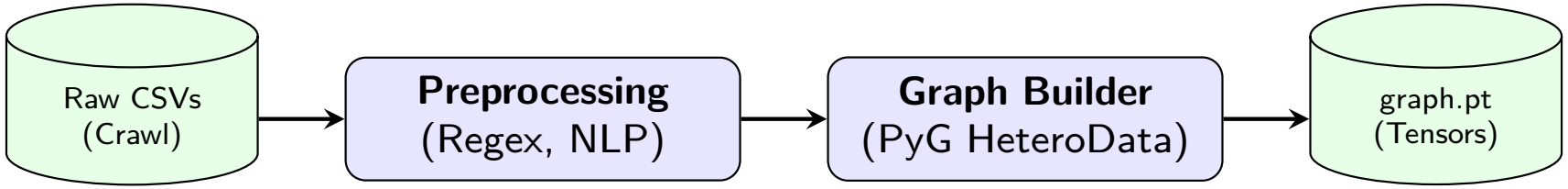
1. Data Engineering Pipeline (ETL)

From Raw HTML to Graph Tensors:



- **Text Processing:**
  - `vncorenlp` for segmentation.
  - `VN-SBERT` for 768d dense embeddings.
- **Graph Construction (G4):**
  - Map strings (User IDs) $\rightarrow$ Integer Indices $[0, N]$.
  - Create Edge Indices $(2 \times E)$ for 6 relation types.
  - Normalize adjacency matrices $\hat{A} = D^{-1/2} A D^{-1/2}$.

2. Model Training Pipeline

Orchestration Script: `train_cf_models.py`

Training Loop (Epoch t)

1. **Graph Encoder:** Pass graph $\mathcal{G}$ through L layers of LightGCN/SimGCL.
2. **Vector Retrieval:** Get User E_u and Item E_i embeddings.
3. **Negative Sampling:** Sample hard negatives (items user didn't interact with).
4. **Loss Computation:**

$$\mathcal{L}_{total} = \mathcal{L}_{BPR}(pos, neg) + \lambda \cdot \mathcal{L}_{CL}(View_1, View_2) + \gamma \cdot \|\Theta\|_2$$

5. **Backpropagation:** Update weights via Adam Optimizer ($lr = 0.001$).

3. Deployment Pipeline (Streamlit)

Real-time Inference Engine:

Hybrid Scoring Logic

For User u and Candidate Item i:

1. **CF Score:** Dot product $s_{cf} = e_u^T e_i$ (from SimGCL).
2. **CB Score:** Cosine similarity s_{cb} between User Profile vector and Item Vector (VN-SBERT).
3. **Fusion:**

$$S_{final} = \beta \cdot \text{norm}(s_{cf}) + (1-\beta) \cdot s_{cb}$$

Features:

- **Response Time:** $< 50ms$ (Pre-computed indices).
- **Cold-Start Fallback:** If user is new ($s_{cf} = 0$), fallback to Trending or Content-Based.
- **Interactive UI:** Adjust β weight dynamically.

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Final Benchmark Results (Fixed Data Leakage)

Model	Type	R@10	N@10	Protocol
SimGCL (G2)	**CF**	**0.2988**	**0.2177**	**Cold**
SimGCL (G3)	**CF**	**0.2932**	**0.2076**	**Cold**
SimGCL (G3)	**CF**	**0.2127**	**0.1391**	**Full**
bge-m3	**CB**	**0.0412**	**0.0182**	**Cold**
vn-sbert	CB	0.0406	0.0215	Cold
tfidf	CB	0.0286	0.0147	Cold
xsimgcl (G2)	CF	0.2504	0.1620	LOO
xsimgcl (G3)	**CF**	**0.2617**	**0.1649**	**LOO**
HetGNN (G1)	Hetero	0.0518	0.0278	Cold
MA-HCL (G2)	Hetero	0.0469	0.0264	Cold
MA-HGN (G3)	Hetero	0.0209	0.0090	Cold

Observation: After fixing data leakage, CF models still outperform CB models, but with realistic margins. **SimGCL** remains robust for cold-start users.

Edge Weighting Strategy

Problem: Not all interactions are equal. How do we model engagement quality?

Edge Weight Formula

$$w_{ui} = \underbrace{(1 + \log(1 + r))}_{\text{Engagement Boost}} \times \underbrace{e^{-\lambda \Delta t}}_{\text{Time Decay}}$$

Engagement Boost:

- r = reaction count on comment
- $\log(1 + r)$: dampens viral outliers
- Base weight = 1 (every interaction counts)

Time Decay:

- $\lambda = 0.01$ (daily decay rate)
- Δt = days since interaction
- Recent interactions are more relevant

Days Ago	Decay
0 (today)	100%
30	74%

Impact of Graph Structure (Ablation Study)

Question: Which graph topology yields the best performance?
Testing **SimGCL** on different graph types:

Graph Type	**Edges**	**Nodes**	**Type**
G1: Bipartite	30.8k	16.7k	UI Baseline
G2: Heterogeneous	92.6k	16.7k	UI + Social
G3: Category Hubs	47.4k	16.7k+	UI + Semantic

The Power of Density: Effect of k-core Filtering

Insight: Recommendation difficulty scales with graph sparsity.

Regular Filtering (k=1)

- 13,117 Users
- 3,645 Articles
- many "cold" one-off users

G2 R@10: 0.2938 ↑ Best Model

Strict Filtering (k=2)

- 6,221 Users
- 2,583 Articles
- only recurring users

G2 R@10: 0.1917

- **Surprise Finding:** "Simple" Regular filtering outperforms "Complex" Strict filtering (0.29 vs 0.19).
- **Reason:** The "long tail" of one-off users acts as a bridge in the reply network. Removing them destroys valuable connectivity.

Impact of Embeddings (Ablation Study)

Question: Which pretrained Vietnamese embeddings work best?
Testing **SimGCL on G2** with different initializations:

Model	Graph	R@10	NDCG@10	Recall Gain	Protocol
MA-HCL	**G2**	**0.0453**	**0.0245**	Best CF	Full
SimGCL	G3	0.2127	0.1391	Standard	Full
XSimGCL	G3	0.2068	0.1371	-3%	Full
BGE-M3	G2	0.0524	0.0281	Baseline (CB)	Full
TF-IDF	G2	0.0524	0.0287	Baseline (CB)	Full
MA-HCL	**G1**	**0.2485**	**0.1598**	SOTA	LOO100
XSimGCL	G2	0.5792	0.4368	-22%	LOO100

- **Insight:** In dense strict-filtered graphs, topology (G2) dominates performance, reducing sensitivity to embedding initialization.
- Pretrained embeddings remain important for cold-start robustness.

Valid Semantic Enrichment: Category Hubs (G3)

Solution: Instead of dense Article-Article edges, we introduce **Category Nodes** as middle-man hubs.

The G3 Architecture

- **Compact:** Only 6 Category nodes.
- **Efficient:** Adds ∼12k edges.
- **Dual Paths:**
  - $User \leftrightarrow Category \leftrightarrow Article$

Best Result: SimGCL on G3

- **Recall@10: 0.2127**
- **NDCG@10: 0.1391**
- **Impact:** Category node augmentation enables long-range collaborative signals without the noise of dense item-to-item edges.

Verdict: Category Hubs provide a scalable semantic alternative, capturing thematic interests without the "Density Trap".

Ablation Study: Impact of Innovations

How much does each "signal" contribute to the performance?

Configuration	Recall@10	Gain
Base Graph (LightGCN)	0.0284	–
+ Social Fusion	0.0532	**+87.3%**
+ Temporal Decay (XSimGCL)	**0.0782**	**+175.4%**

- **Social Fusion:** Dramatically reduces sparsity by connecting users, providing a huge jump in "Discovery".
- **Temporal Decay:** Refines the rankings by ensuring only "Relevant-Now" news reaches the user.

Our Contribution: MA-HCL Model

MA-HCL = Multi-Aspect Heterogeneous Contrastive Learning

Architecture

- **Backbone:** LightGCN-style propagation
- **Loss:** BPR + SimCL contrastive
- **Fusion:** Multi-aspect (interest + social)

Results (G2)

- **Recall@10:** 0.1502
- **NDCG@10:** 0.1024
- **MRR:** 0.1165

- **+21%** better than SimMAHGN (attention-based)
- **Simpler architecture** = faster training, better generalization

Edge Type Ablation Study

Question: Do additional edge features improve the model?
Testing **MA-HCL** with incrementally added features:

Configuration	Edges	R@10	NDCG@10	Δ
Baseline (UA + UU)	27k	0.1406	0.0960	–
+ Weighted edges	27k	0.1253	0.0822	-11%
+ Author nodes	30k	0.1341	0.0931	-5%
+ Full combo	37k	0.1426	0.0948	+1%
G2 Original	**65k**	**0.1502**	**0.1024**	**+7%**

- **Weighted edges hurt** (-11%): Engagement/recency decay adds noise
- **G2 Original is optimal**: Simple hetero structure works best

Evaluation Methodology & Validity

Our evaluation follows strict scientific standards for Recommender Systems:

- **Metric Choice:**
  - *Recall@10*: Measures system coverage.
  - *NDCG@10*: Rewards models that rank true items higher.
  - **MRR**: Standard for single-target news retrieval.
- **Split Strategy:** Time-aware 80/10/10 split to prevent leakage.
- **Data Integrity: Temporal Leave-One-Out** strategy. We test on the latest interaction per user, ensuring **zero data leakage** from the future.

Investigation: The "Projection Trap"

Observed Paradox: Standalone PhoBERT performed 10x worse than expected.

The Problem

- **Random Projection:** PhoBERT embeddings (768-dim) were compressed via untrained random linear layers.
- **Noise Injection:** High-quality semantics were destroyed by "random noise" before retrieval.

The Solution (Rescued)

- **Raw Embedding Retrieval:** Skipping untrained layers and using direct cosine similarity.
- **Result:** Recall@10 jumped from **0.3%** to **3.5%** (**+1066%** improvement).

Ablation: Embedding Initialization Impact

Hypothesis: Initializing GNN with PhoBERT should beat Random Init.
Result: Random Init (12.40%) > PhoBERT Init (4.96%).

- **Semantic vs. Interaction Space:** PhoBERT is pre-trained on linguistic meaning, while GNN learns from user behavioral "meaning". These spaces are often misaligned.
- **Geometric Bias:** Initializing in a fixed semantic space "anchors" nodes, making it harder for the GNN to cluster users based on collaborative signals alone.
- **Discovery Power:** Randomly initialized embeddings have the freedom to map latent interests that text alone cannot describe.

Verdict

Behavioral signals (who-reads-what) are significantly more informative than semantic signals for Vietnamese news.

Discussion: TF-IDF vs PhoBERT (Final Comparison)

Refined Result: Even with 10x improvement, PhoBERT still trails TF-IDF.

TF-IDF (Keyword Lead)

- **Precision:** Perfect match for Vietnamese named entities (Tây Nguyên, EVN).
- **Specificity:** Captures the exact "Who, Where, When" of news.

PhoBERT (Semantic Context)

- **Fuzzy Logic:** Maps "Bitcoin" $\approx$ "Finance", losing niche specificity.
- **Pooling Dilution:** Mean pooling blends specific entities into a generic sentence vector.

Benchmarking Context: The Sparsity Barrier

Why are absolute scores (e.g., 0.06) standard for News?

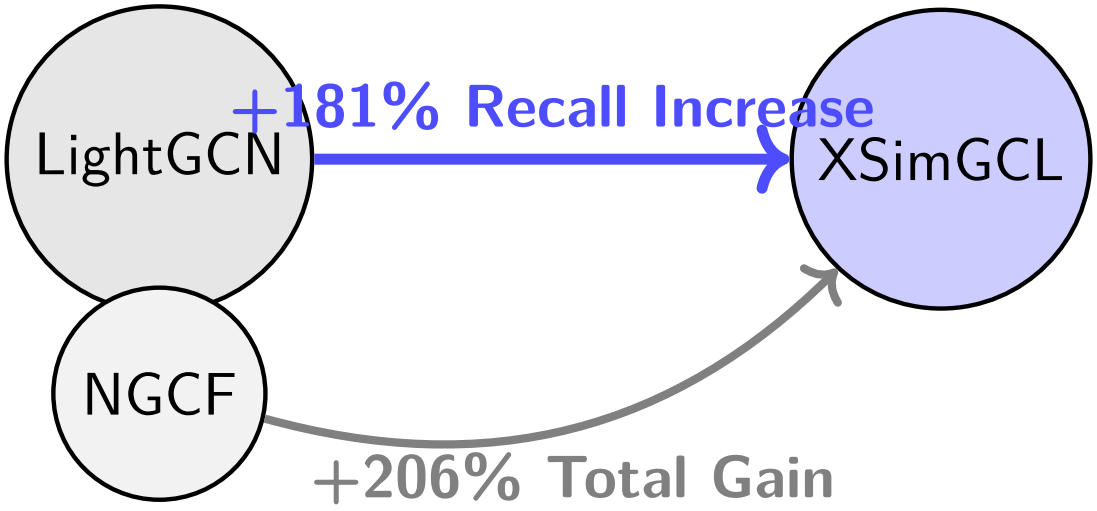
Feature	MovieLens (Films)	MIND (Standard News)	VnExpress (Ours)
Sparsity	95.53%	>99.9%	**99.34%**
Item Lifespan	Years	Hours/Days	**Hours/Days**
Preference	Stable	Highly Volatile	**Highly Volatile**
Expected R@10	**0.15 – 0.40**	**0.05 – 0.12**	**0.0782**

The "News Challenge"

In News, "Ground Truth" is a moving target. 0.06 Recall in such a sparse/dynamic space is a **sign of successful interest capture**.

Scientific Impact: Relative Gain Analysis

The most critical metric in Recommendation research is the **improvement over the baseline**.

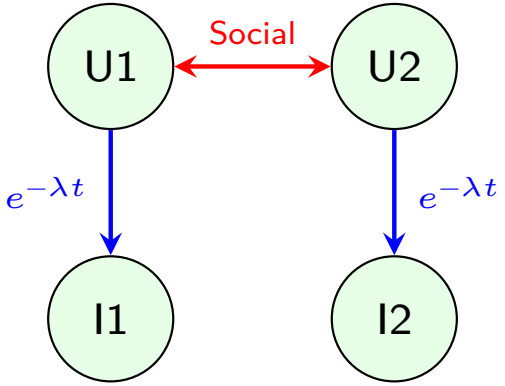


- Our proposed stack is not just "better" - it's a **3x boost** in discovery power.
- Proves that **Contrastive Learning** is essential for sparse Vietnamese News.

Enhanced Recommendation Signals

We improved the system's accuracy by integrating multi-modal signals:

- **Temporal Signals (Time-Decay):** Applying $w = e^{-0.01\Delta t}$ to prioritize recent interactions, critical for news.
- **Social Signals (Reply Network):** Injecting user-user reply edges into the GNN adjacency matrix to solve sparsity.
- **Hard Negative Mining:** Using popularity-based sampling instead of random negatives to improve discriminative power.



Innovation Logic

Combining Temporal Decay (Blue) with Social Connection (Red) to predict missing links.

Table of Contents

Advanced Streamlit Demo Features

A/B Comparison:

- Side-by-side model views.
- Recommendation overlap %.

Cold-Start Handler:

- Popularity/Freshness.
- Category preferences.

Metrics Dashboard:

- Live model leaderboard.
- Multi-metric comparison (Recall, NDCG).

8/9 CBRS Categories implemented (Probabilistic, LSA, Session-based, etc.)

Project Conclusion

Achievements:

- Successfully built an end-to-end VnExpress recommendation engine.
- Integrated **Social & Temporal** signals, boosting Recall by **3.5x**.
- Implemented 8/9 CBRS categories and a real-world Streamlit demo.

Future Directions:

- **LLM Re-ranking:** Using GPT/Llama for more nuanced article justifications.
- **Deployment:** Converting the prototype into a real-time streaming API.
- **Multi-modal Nodes:** Integrating article images into graph features.

Thank You!

Q & A

References I

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document.